# TAMIL NADU TEACHERS BENCHMARK PORTAL

**OFFICIAL TNTET EXAM LEARNING & BENCHMARK EXAMINATION HUB • TNTEACHERS.IN**

**PRODUCTION V2.6 • ENTERPRISE**
Host: 66.116.250.237

**Live URL:** https://tnteachers.in

**Executive Project Delivery & Complete Technical Specification Handover**
Comprehensive milestone handover document detailing full platform architecture, 9 core modules, 5-stage sequential learning engine, Razorpay monetization, automated coupon engine, daily practice tests, and cloud VPS infrastructure.

**DATE:** October 2026
**STATUS:** Production Live & Verified
**HOST:** 66.116.250.237 (Ubuntu VPS)
**DOMAIN:** tnteachers.in (SSL Active)

## 1. Executive Summary & Delivery Scope

The **Tamil Nadu Teachers Benchmark Portal (tnteachers.in)** is a premier, full-stack educational and competitive examination platform specifically engineered for **Tamil Nadu Teacher Eligibility Test (TNTET)** Paper 1 & Paper 2 candidates, B.Ed / D.T.Ed graduates, government school educators, and competitive scholarship aspirants. The platform delivers structured digital textbook repositories, high-yield chapter notes, video classroom explanations, interactive practice engines, and timed benchmark model examinations with an automated 80% passing certificate.

| | | | | |
|---|---|---|---|---|
| **146+ LESSONS**<br>Classes 1–12 Full Syllabus | **10,000+ QUESTIONS**<br>Bilingual Tamil & English Bank | **5 SEQUENTIAL STAGES**<br>Book • Notes • Video • Practice • Exam | **RAZORPAY + COUPONS**<br>Automated Discounts & Checkout | **PWA MOBILE APP**<br>Installable on Android & iOS |

## 2. The 9 Core Educational & Examination Modules

| MODULE NAME | ACCESS LEVEL | TARGET AUDIENCE & FUNCTIONAL CAPABILITIES |
|---|---|---|
| **1. TET - 1 (Paper 1)** | **PREMIUM COMBO** | Classes 1 to 8 SGT (Secondary Grade Teacher) preparation. Covers Child Development & Pedagogy, Tamil, English, Mathematics, and EVS. Features all 5 study stages with automatic 80% passing certification. |
| **2. TET - 2 (Paper 2)** | **PREMIUM COMBO** | Classes 6 to 12 Graduate Teacher (BT Assistant) preparation across Science, Mathematics, and Social Science streams. Incorporates dual question banks (Type 1 Primary & Type 2 Alternate) for authentic testing. |
| **3. Online Live Class Hub** | **PREMIUM COMBO** | Interactive live virtual classroom integrating Google Meet, Zoom, YouTube Live, and Webex. Includes live scheduling countdowns, recorded archives, and a real-time pulsing **LIVE NOW** alert indicator. |
| **4. TN Textbooks Explorer** | **FREE PUBLIC** | Complete Classes 1 to 12 Tamil Nadu State Board (Samacheer Kalvi) digital curriculum repository. Dynamic multi-filtering by Class, Subject, Medium (Tamil/English), and Term. In-browser responsive PDF reader. |
| **5. Minnal Educational News** | **FREE PUBLIC** | Dedicated mobile feed (**/news**) broadcasting real-time educational flash news, TRB notifications, hall ticket release alerts, syllabus revisions, and School Education Government Orders (G.O.). |
| **6. Tamil Reading Mastery** | **FREE PUBLIC** | Tamil literary enrichment module offering pronunciation drills, grammatical treatises (Ilakkanam, Ilakkiyam, Urainadai), and reading comprehension exercises tailored for teacher candidates. |
| **7. English Reading Mastery** | **FREE PUBLIC** | Grammar drills, phonetics, reading comprehension passages, and pedagogical English vocabulary builder formulated to build English language proficiency for Paper 1 and Paper 2 candidates. |
| **8. NMMS Exam Preparation** | **FREE PUBLIC** | National Means-cum-Merit Scholarship portal for Class 8 students with Mental Ability Test (MAT) and Scholastic Aptitude Test (SAT) practice questions, explanations, and model question papers. |
| **9. TREAST / TRUST Exam Hub** | **FREE PUBLIC** | Tamil Nadu Rural Students Talent Search Examination portal offering subject-wise practice tests, previous year papers, and model question papers for government school rural scholars. |

## 3. The 5-Stage Sequential Learning & Examination Engine

To ensure thorough pedagogical mastery and prevent shallow memorization, every curriculum lesson enforces an innovative **5-Stage Sequential Study Pipeline**. Candidates unlock subsequent stages and future lessons only upon completing prerequisite steps and achieving qualifying benchmarks:

| STAGE & STEP | MODULE TITLE | FUNCTIONAL SPECIFICATION & CANDIDATE WORKFLOW |
|---|---|---|
| **Stage 1**<br>**BOOK** | **Lesson Textbook PDF**<br>Curriculum Textbook PDF | Candidates read the complete government textbook chapter directly in-browser. Integrated responsive PDF reader with page controls, zoom, and text search. Admin can replace or update the PDF anytime. |
| **Stage 2**<br>**NOTES** | **Short Notes & Summary**<br>High-Yield Revision PDF | High-yield revision points, summary bullet charts, and important memory anchors curated for fast review before tackling practice questions. Admin can upload replacement notes PDF. |
| **Stage 3**<br>**VIDEO** | **Video Lecture Class**<br>Embedded Video Explanation | Embedded video tutorial explaining core concepts, pedagogical nuances, and memory shortcuts. Features responsive layout with fullscreen theater support and YouTube embed player integration. |
| **Stage 4**<br>**PRACTICE** | **Practice Test Engine**<br>Type 1 Practice MCQs | Interactive practice engine with instant answer validation, correct answer reveals, pedagogical theory explanations, and untimed learning mode to reinforce conceptual clarity. |
| **Stage 5**<br>**FINAL EXAM** | **Online Timed Test (≥80%)**<br>Type 2 Benchmark Exam | Formal timed examination under real TNTET conditions. Requires scoring at least **80% (Pass Benchmark)** to certify lesson completion and unlock the next lesson in the curriculum. Automatically locks back to Restudy Mode if score is < 80%. |

### Key Progression & Unlocking Rules:

- **Sequential Curriculum Lock:** Lessons must be cleared in order (Term 1 $\rightarrow$ Term 2 $\rightarrow$ Term 3; Lesson 1 $\rightarrow$ Lesson 2). A lesson only unlocks when the previous lesson achieves a $\ge 80\%$ qualifying score.
- **Dual Question Banks:** Stage 4 utilizes Type 1 Primary questions, while Stage 5 utilizes Type 2 Alternate questions. If a candidate scores below 80%, the retake exam draws fresh unmemorized questions.
- **Daily Schedule Time-Lock:** By default, Stage 5 examinations unlock automatically at **8:00 PM IST** daily. Administrators can override this with the **Force Unlock Switch (Open Immediately)** to unlock tests for students right away.

## 4. Daily Practice Test System & Custom Stage Selection

In addition to sequential curriculum progression, the platform includes a high-impact **Daily Practice Module** designed to cultivate daily study streaks and deliver targeted mock tests directly onto the student dashboard:

- **Dashboard Highlighting:** Featured at the top of the student home screen under *'Today's Daily Practice Tests (Daily Model Exams)'* for immediate one-click test taking.
- **Custom Stage Cherry-Picking:** While full curriculum lessons enforce all 5 stages, the Daily Practice Admin tool allows selecting **any custom subset of the 5 stages** for that day (e.g., Stage 1 Notes + Stage 4 Practice + Stage 5 Exam only).
- **Timed Examination Engine:** Candidates take tests with an active countdown timer, auto-submission on expiration, and instant performance analysis against the 80% passing benchmark.
- **Admin Verification & Export:** Admins can preview questions using **View Quiz** or download the test using **Download PDF** for offline printing and content verification.

## 5. Razorpay Monetization & Automated Coupon Engine

A full-fledged commercial monetization system enables turnkey course purchases, lead capture, and promotional discount code implementation:

| COMMERCIAL CAPABILITY | TECHNICAL SPECIFICATION & AUTOMATION BEHAVIOR |
|---|---|
| **All-In-One Combo Pass** | Enrolling or paying for either TET 1 or TET 2 automatically unlocks all 3 premium modules simultaneously (TET 1 + TET 2 + Online Live Class). |
| **Course Pricing Controls** | Admin configurable **Original MRP** (e.g. Rs. 2,499), **Course Selling Price** (e.g. Rs. 999), and **Default Discount** (e.g. 60%) displayed with live preview badges. |
| **Razorpay Payment Gateway** | Client-side dynamic gateway checkout (checkout.js) with pre-filled candidate details. Server-side HMAC-SHA256 signature verification guarantees 100% cryptographic payment security. |
| **Automated Coupon Engine** | Create custom codes or auto-generate promo codes (e.g. TET60, DIWALI2026). Supports Percentage (%) or Flat (Rs.) discounts, minimum order amount, and maximum discount caps. |
| **Schedule Automation** | Coupons include automated start date/time (auto-enable) and end date/time (auto-disable/expire). System automatically transitions status (Active, Scheduled, Expired, Exhausted, Disabled). |
| **Redemption Limit Tracking** | Enforce maximum allowed redemptions (e.g. First 100 students). Times-used counter increments atomically upon successful payment verification. |
| **Admin Free Enrollment Override** | Administrators can manually enroll any candidate for free directly from the Admin Panel, bypassing the payment gateway with authority audit notes. |

## 6. High-Fidelity PDF Quiz Export & Formatting Engine

A dedicated print-ready PDF export engine (pdfExport.js) allows students and administrators to generate beautifully typeset, aligned quiz papers:

- **Two-Column Question Stem Layout:** Clean separation of question numbering and question content, eliminating uneven margins and preventing ragged gaps.
- **Adaptive Option Grid:** Automatically renders options in a 2-column grid for short answers ($\le 34$ chars) or full-width rows for long statements.
- **Linguistic Text Sanitization:** Universal regex cleaning that removes duplicate numbering (1., (A)), InDesign metadata (.indd), timestamps, running headers, and fixes split Tamil pulli/vowel glyphs.
- **Page Break Avoidance:** Individual questions are guarded with CSS break-inside: avoid to prevent questions from splitting across page boundaries.

## 7. Super-Admin Management Center (/admin)

The protected administration back-office delivers complete operational control over all website data, content, learners, and system configurations across 11 dedicated management tabs:

| ADMIN TAB | OPERATIONAL CAPABILITIES & WORKFLOWS |
|---|---|
| **1. Overview & Stats** | Real-time key metrics: Total registered students, enrolled paid candidates, lesson modules published, question count, and database health. |
| **2. Enrollments CRM** | Search, filter, and monitor all student purchase leads, payment IDs, order IDs, and status. Includes **+ Enroll Candidate Free** button for administrative access grants. |
| **3. Daily Practice CMS** | Create daily class-wise tests with custom stage selection, attach study PDFs, set question counts, test duration, and download formatted quiz verification PDFs. |
| **4. Lesson Modules CMS** | Upload new lessons, view complete catalog, edit core metadata, and launch the **Edit Lesson Module & Schedule** modal with direct PDF replacements and stage time-locks. |
| **5. Quiz Management** | Manage Type 1 and Type 2 question banks. Incorporates Google Gemini Generative AI question generator, single-question regenerator, and bulk manual question editors. |
| **6. Live Meetings CMS** | Schedule and broadcast live virtual classes with Google Meet, Zoom, or YouTube links. Controls active meeting status and triggers the student 'LIVE NOW' banner. |
| **7. Materials CMS** | Curriculum library manager for Class 1 to 12 textbook PDFs. Upload, organize by subject and medium, preview documents, and delete old syllabus files. |
| **8. Settings & Pricing** | Configure Razorpay API Keys, set Course MRP & Selling Price, toggle website-wide student Material PDF & Quiz PDF download permissions, and manage Promo Coupons. |
| **9. Announcements** | Publish breaking educational ticker announcements displayed across student dashboards with priority badges and expiry controls. |
| **10. Learning Containers** | Customize and reorder all 9 learning containers on the student dashboard, configure access rules, and modify badge labels. |
| **11. VPS Host Monitor** | Server operations monitor: tracks host CPU, memory, disk usage, active systemd service status (tet-backend.service), and displays live journal logs. |

### Enhanced Edit Lesson Module & Schedule Modal:

- **Comprehensive 5-Stage Configuration:** Direct file upload/replacement for Stage 1 Textbook PDF and Stage 2 Short Notes PDF, Video URL input for Stage 3, and unlock time-locks & force-open overrides for Stage 4 & 5.
- **Active Stage Inclusion Selector:** Checkboxes allowing administrators to selectively enable/disable specific stages for any lesson.
- **Release Date Scheduling:** Future release date & time calendar lock, with quick presets (Immediate Release, Tonight 8 PM, Tomorrow 6 AM).

## 8. Technical Stack & Infrastructure Specifications

| LAYER | TECHNOLOGY & VERSION | ARCHITECTURAL ROLE & SPECIFICATION |
|---|---|---|
| **Frontend SPA** | React 19.2 + Vite 8.2 | Component-based Single Page Application with fast ESM bundling and minimal production footprint. |
| **Styling & Icons** | Tailwind CSS v4 + Lucide | Utility-first design system optimized for mobile viewports, safe-area padding, and high-DPI displays. |
| **Backend Framework** | FastAPI 0.110 (Python 3.11) | High-performance asynchronous REST API framework with Pydantic v2 data validation schemas. |
| **ASGI Server** | Uvicorn 0.28 (2 Workers) | Production multi-worker event loop managed by Linux systemd (tet-backend.service). |
| **Relational Database** | MariaDB / MySQL 8.0 | Enterprise relational database with SQLAlchemy 2.0 ORM, connection pooling, and automatic migrations. |
| **Cloud VPS Host** | Ubuntu Linux 22.04 LTS | Dedicated cloud instance (IP: **66.116.250.237**) with Nginx reverse proxy and HTTP/2. |
| **SSL & Security** | Let's Encrypt Certbot | Automated TLS 1.3 encryption, CORS origins whitelisting, and HMAC-SHA256 signature verification. |
| **PWA Support** | Service Worker + Manifest | Installable on Android and iOS home screens with offline asset caching and full-screen experience. |

## 9. Key REST API Endpoints Specification

| METHOD | ENDPOINT ROUTE | DESCRIPTION & ACCESS SCOPE |
|---|---|---|
| **GET** | /api/public-settings | Returns public portal settings: download permissions, Razorpay key, course price, and discount %. |
| **GET** | /api/learning/lesson/{id} | Retrieves lesson textbook PDF, short notes, video URL, stage progress, and time-lock states. |
| **POST** | /api/learning/lesson/{id}/submit-test | Evaluates student Stage 5 exam submission, computes percentage, and enforces 80% passing rule. |
| **POST** | /api/coupons/validate | Validates coupon code, checks date automation and usage limits, and returns discounted amount. |
| **POST** | /api/enrollments/register | Captures student lead details and creates pending order with optional coupon code applied. |
| **POST** | /api/enrollments/verify-payment | Verifies Razorpay HMAC signature, activates course access, and increments coupon redemptions. |
| **GET** | /api/daily-practice | Retrieves active daily practice mock tests configured with custom stages for student dashboard. |
| **GET** | /api/meetings/active | Returns ongoing live classroom meetings, triggering the student portal 'LIVE NOW' banner. |
| **GET** | /api/admin/lessons | Admin endpoint listing all 146+ curriculum lessons with 5-stage status, question counts, and locks. |
| **PUT** | /api/admin/lessons/{id} | Updates lesson metadata, textbook PDF URL, short notes URL, video URL, and Stage 4/5 schedules. |
| **POST** | /api/admin/lessons/{id}/upload-pdf | Uploads and replaces Stage 1 Lesson Textbook PDF file directly on the production server. |
| **POST** | /api/admin/lessons/{id}/upload-short-notes | Uploads and replaces Stage 2 Short Notes PDF file directly on the production server. |
| **POST** | /api/admin/coupons | Creates a new promo coupon with percentage/flat discount, date range automation, and usage limits. |
| **POST** | /api/admin/enrollments/manual | Administrative override to grant 100% free full access to any student without payment. |

## 10. Final Project Handover, Verification & Sign-Off

All requested platform enhancements, architectural modifications, coupon engines, stage configuration controls, PDF quiz alignment fixes, and production cloud deployments have been successfully concluded, verified, and placed into live production.

| DELIVERABLE / MILESTONE ITEM | VERIFICATION RESULT | EVIDENCE & LIVE BEHAVIOR |
|---|---|---|
| **1. All 9 Educational Modules** | **VERIFIED LIVE** | TET 1, TET 2, Live Classes, Textbooks, Minnal News, Tamil, English, NMMS, TRUST active. |
| **2. 5-Stage Sequential Engine** | **VERIFIED LIVE** | Book, Notes, Video, Practice, and 80% Timed Final Exam progression verified on curriculum. |
| **3. Daily Practice System** | **VERIFIED LIVE** | Custom stage selection, dashboard cards, timed test modal, and admin PDF export active. |
| **4. Razorpay + Coupon Engine** | **VERIFIED LIVE** | Pricing (Rs. 2499 / Rs. 999), promo codes, date automation, limits, and verification active. |
| **5. PDF Quiz Print Alignment** | **VERIFIED LIVE** | 2-column numbering grid, adaptive option layout, and InDesign/.indd metadata stripped. |
| **6. Edit Lesson 5-Stage CMS** | **VERIFIED LIVE** | All 5 stages editable with direct PDF replacement upload and schedule time-locks. |
| **7. Cloud VPS & SSL Operations** | **VERIFIED LIVE** | Host 66.116.250.237, Uvicorn service active, MariaDB active, Nginx SSL certificate verified. |

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|---|---|
| **PROJECT TITLE:** | Tamil Nadu Teachers Benchmark Portal (TNTET Learning & Exam Platform) |
| **LIVE WEB PORTAL:** | https://tnteachers.in \| **Public IP:** 66.116.250.237 |
| **SOURCE REPOSITORY:** | GitHub / tho-mas12 / tet-color (Branch: main) |
| **RELEASE VERSION:** | Production Release v2.6 Enterprise (October 2026) |
| **DELIVERABLE STATUS:** | **APPROVED, VERIFIED & 100% PRODUCTION OPERATIONAL** |

*TN Teachers Benchmark Portal Technical Specification • Prepared for Project Stakeholders & System Verification • © 2026 Confidential & Proprietary*